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Student Number



Barker
College

Biology

2021

TRIAL HIGHER SCHOOL
CERTIFICATE EXAMINATION

ANSWER SHEET

Staff Involved:

AM THURSDAY 26th AUGUST

- TKB • KJH
- SLC • RYF
- DGG • KIP

110 copies

Section I - Multiple Choice

Write your Student Number at the top of this page

Choose the best response and fill in the response oval completely

Start
Here

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|-----|-----|-----|-----|-----|----|-----|-----|-----|-----|
| → 1 | A ○ | B ○ | C ○ | D ○ | 11 | A ○ | B ○ | C ○ | D ○ |
| 2 | A ○ | B ○ | C ○ | D ○ | 12 | A ○ | B ○ | C ○ | D ○ |
| 3 | A ○ | B ○ | C ○ | D ○ | 13 | A ○ | B ○ | C ○ | D ○ |
| 4 | A ○ | B ○ | C ○ | D ○ | 14 | A ○ | B ○ | C ○ | D ○ |
| 5 | A ○ | B ○ | C ○ | D ○ | 15 | A ○ | B ○ | C ○ | D ○ |
| 6 | A ○ | B ○ | C ○ | D ○ | 16 | A ○ | B ○ | C ○ | D ○ |
| 7 | A ○ | B ○ | C ○ | D ○ | 17 | A ○ | B ○ | C ○ | D ○ |
| 8 | A ○ | B ○ | C ○ | D ○ | 18 | A ○ | B ○ | C ○ | D ○ |
| 9 | A ○ | B ○ | C ○ | D ○ | 19 | A ○ | B ○ | C ○ | D ○ |
| 10 | A ○ | B ○ | C ○ | D ○ | 20 | A ○ | B ○ | C ○ | D ○ |

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Student Number



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**TRIAL HIGHER SCHOOL
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Biology

Staff Involved:

AM THURSDAY 26th AUGUST

- TKB • KJH
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General

Instructions:

- Reading time - 5 minutes
- Working time - 3 hours
- Write your Student Number at the top of every page of your lined paper.
- Write using black pen
- Draw diagrams using pencil
- For questions in Section II, show all relevant working in questions involving calculations
- NESA approved calculators may be used

Total marks:

100

Section I - 20 marks (pages 2- 12)

- Attempt Questions 1 - 20
- Allow about 35 minutes for this section

Section II - 80 marks (pages 13 - 31)

- Attempt Questions 21 - 31
- Allow about 2 hours and 25 minutes for this section

Section I

20 marks

Attempt Questions 1 – 20

Allow about 35 minutes for this part

Use Canvas to select your preferred response for Questions 1 – 20

Select the alternative A, B, C or D that best answers the question.

Do not fill in the response oval for the 2021 Trial paper

Sample $2 + 4 =$ A 2 B 6 C 8 D 9
 A ☐ B ☐ C ☐ D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ● B ☒ C ☐ D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate this by writing the word *correct* and drawing an arrow as follows.

A ● B ☒ C ☐ D ☐
 Correct
 ↙

-
1. An outbreak of a serious new strain of influenza occurs on a cruise ship at sea. What would be the best method of preventing the influenza from spreading to populations on land?
 - A. Keep all people on the ship until everyone has recovered.
 - B. Send crew members ashore to obtain antiviral medication.
 - C. Disinfect all eating and recreational areas on the ship.
 - D. Bring in medical personnel to vaccinate people on the ship.

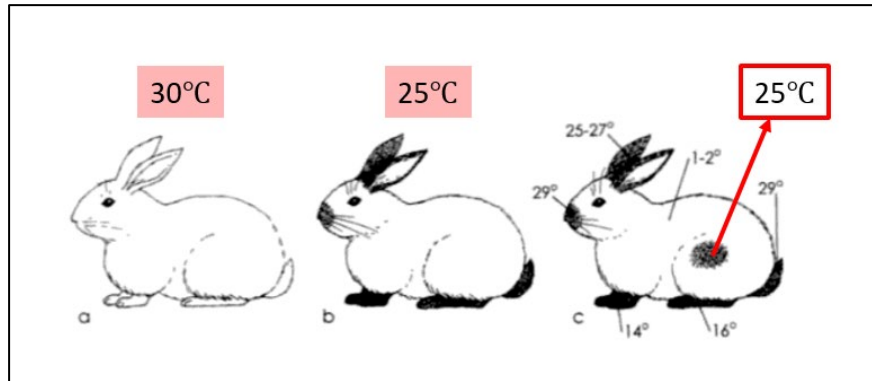
 2. Which of the following statements about proteins is true?
 - A. Proteins are the building blocks of amino acids.
 - B. Proteins are small, simple molecules.
 - C. Proteins provide structure and support for cells.
 - D. Proteins are types of polysaccharides.

3. Which of the following will reduce genetic differences between populations?
- A. Gene flow
 - B. Mutation
 - C. Sexual selection
 - D. Genetic drift
4. A student disinfected their workbench with alcohol wipes before conducting an experiment to test for microbes in water.

Why did the student disinfect their workbench?

- A. The student did not want to become infected with microbes from the workbench.
 - B. The alcohol wipes help prevent the experiment becoming contaminated by microbes on the workbench.
 - C. To improve the reliability of the investigation.
 - D. The microbes in the water may contaminate the workbench.
5. Which of the following best describes artificial selection?
- A. Intentional reproduction of individuals with desirable characteristics that changes the frequency of alleles in the selected line.
 - B. Intentional reproduction of individuals with desirable characteristics that does not change the frequency of alleles in the selected line.
 - C. Unintentional reproduction of individuals with desirable characteristics that changes the frequency of alleles in the selected line.
 - D. Unintentional reproduction of individuals with desirable characteristics that does not change the frequency of alleles in the selected line

6. Fur colour in rabbits is determined by four alleles of the *C* gene. Himalayan rabbits are homozygous for the c^h allele of this gene. This allele produces an enzyme that is used in a pathway that results in black fur. However, this enzyme is inactivated if the skin temperature exceeds 20 °C, resulting in white fur.



Fur colour in Himalayan rabbits as described above is an example of

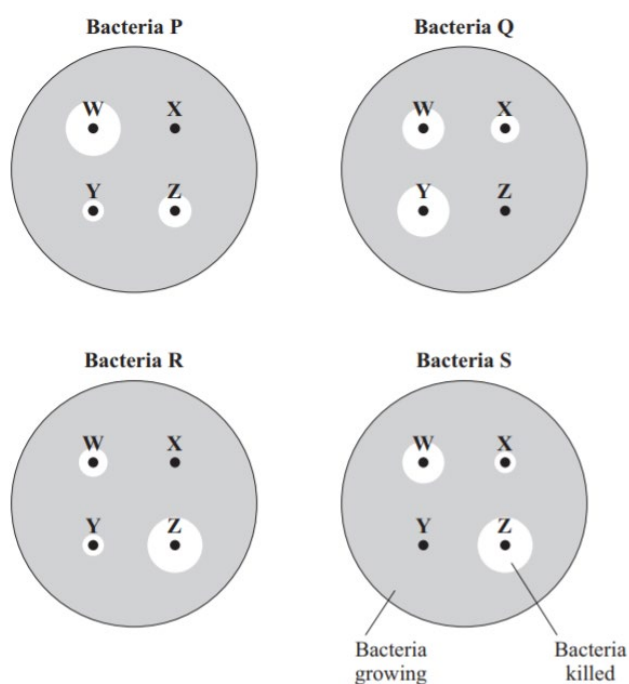
- A. co-dominant inheritance
- B. a recessive mutation
- C. an interaction between genes
- D. gene - environment interaction

7. A group of students investigated how effective four different antibiotics were at killing four different bacteria.

In this investigation:

- the four bacteria, P, Q, R and S, were spread on separate dishes containing nutrient agar.
- paper discs were dipped into the four different antibiotics, W, X, Y and Z .
- one of each disc was placed onto the agar in each of the four dishes.

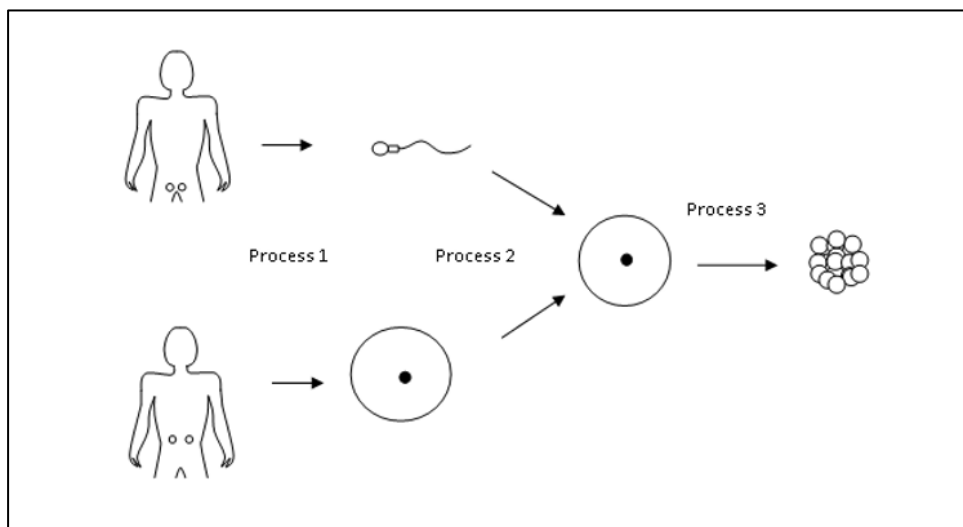
The diagram below shows the appearance of the dishes after three days. If the antibiotic kills the bacteria there is a clear space around the disc. The larger the clear space, the more effective the antibiotic is.



Which antibiotic is most effective against the widest range of bacteria?

- A. W
- B. X
- C. Y
- D. Z

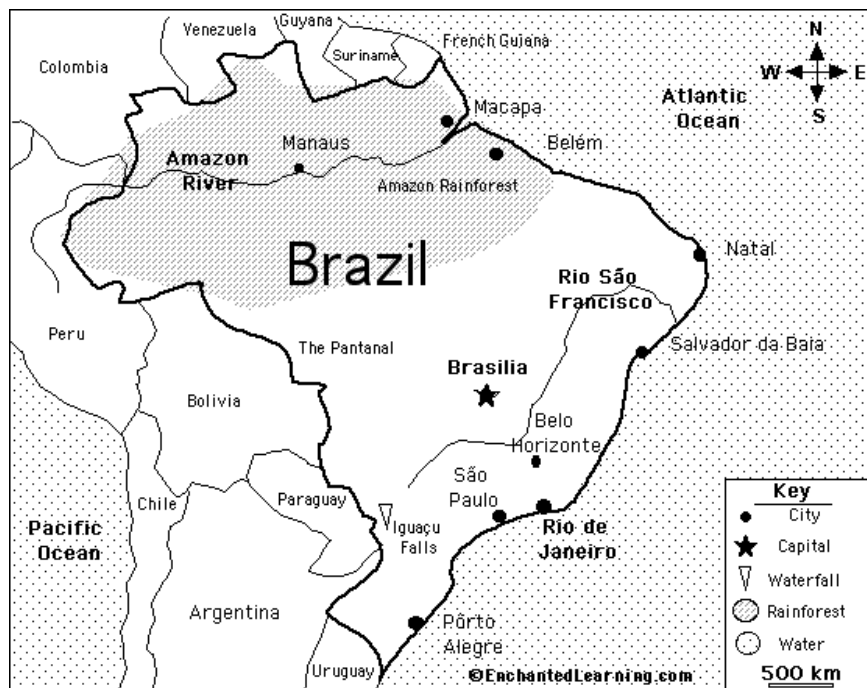
8. In some plant species, if a red flowered plant is crossed with a white flowered plant, they produce a pink hybrid. What type of inheritance is this an example of?
- Recessive inheritance
 - Autosomal dominant
 - Incomplete dominance
 - Autosomal co-dominance
9. Which one of the following statements describes the process of crossing over in meiosis?
- Sister chromatids exchange DNA
 - DNA is exchanged between chromosomes that are not homologous
 - Genes are exchanged between chromatids of the same chromosome
 - Genes are exchanged between homologous chromosomes
10. Question 10 refers to the diagram below.



Choose the row in the table below which correctly describes processes 1 – 3 in the diagram.

	<i>Chromosome number reduced</i>	<i>Chromosome number maintained</i>	<i>Chromosome number increased</i>
A.	Process 1	Process 2	Process 3
B.	Process 3	Process 1	Process 2
C.	Process 2	Process 3	Process 1
D.	Process 1	Process 3	Process 2

11. In eukaryotic cells:
- A. exons are removed from mRNA molecules after transcription.
 - B. exons are added to mRNA molecules after transcription.
 - C. introns are removed from mRNA molecules after transcription.
 - D. introns are added to mRNA molecules after transcription.
12. Yellow fever is a viral disease that is transmitted primarily by mosquitos. An outbreak of yellow fever was reported to have occurred in an area of Brazil in January 2017. The outbreak was reported to be spreading to other areas in Brazil.



Which of the following is a correct statement about this outbreak of yellow fever?

- A. This outbreak of yellow fever is considered to be a pandemic.
- B. Infected individuals who travel to other areas of Brazil will spread the disease.
- C. This outbreak of yellow fever is occurring in populations with high vaccination rates.
- D. Elimination of mosquito breeding sites will reduce the number of individuals affected.

13. A pathogen is identified as being adapted to moist conditions and having the gastrointestinal tract as its route of entry and route of exit from the body.

The most likely mode of transmission for this pathogen is

- A. skin to skin contact
 - B. coughing or sneezing
 - C. contaminated water supply
 - D. transmission of infected blood products
14. The information shown is a summary of the mode of transmission of a range of infectious diseases.

Transmission of infectious disease	
HIV is a virus that is spread through sexual activity.	
Conjunctivitis is an eye disease that is spread by a person touching their infected eye and then touching the eye or face of another person.	
SARS is an infectious disease that can be transmitted by sharing utensils.	
Measles can be transmitted by one person entering a room after an infectious person has left the room.	

Which of the following gives the correct classifications of transmission of the diseases listed above?

	<i>Direct Transmission</i>	<i>Indirect Transmission</i>
A	SARS, measles	HIV, conjunctivitis
B	SARS, HIV	conjunctivitis, measles
C	SARS	HIV, conjunctivitis, measles
D	HIV, conjunctivitis	SARS, measles

15. Consider the following statement:

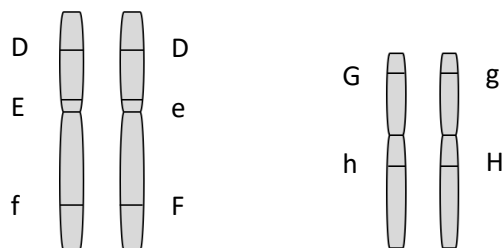
“Once a pathogen has penetrated a cell, the immune system cannot fight it.”

Which of the following is **CORRECT**?

- A. The statement is false as antibodies can penetrate cells to fight pathogens.
 - B. The statement is true as antibodies can only destroy pathogen cells while they are in the bloodstream or extracellular fluid.
 - C. The statement is false as cytotoxic T cells can destroy infected host cells.
 - D. The statement is true as cytotoxic T cells can only destroy pathogen cells while they are in the bloodstream or extracellular fluid.
16. A pathogen infecting a plant was found to have membrane bound organelles, a chitinous cell wall and to feed on the plant tissues.

It can best be classified as a:

- A. protozoan
 - B. plant
 - C. bacterium
 - D. fungus
17. The diagram below shows two pairs of homologous chromosomes of an animal and the arrangement of the alleles of 5 genes on them.



Assuming that no crossing over occurs, which row shows possible genotypes of gametes produced by this animal.

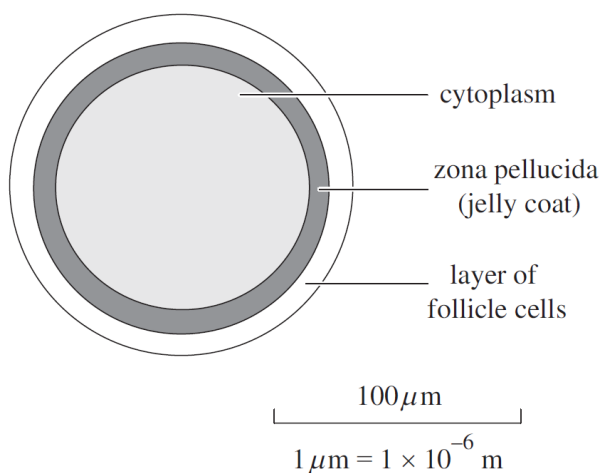
- A. DEf/GH; DEf/gH; DeF/Gh; DeF/gH
- B. DeF/gH; DEF/gH; DeF/GH; DeF/gH
- C. DEf/Gh; DEf/gH; DeF/Gh; DeF/gH
- D. Def/GH; DEf/gH; DeF/Gh; DeF/gH

18. Oestrus signifies when a cow ‘comes into heat’ and is sexually responsive. The diagram compares artificial insemination with natural mating and shows the link between the likelihood of conception and the onset of oestrus.

time after cow comes into heat (hours)	0	6	12	18	24	30
artificial insemination	too early	good	optimal	good	too late	
natural mating	too early	optimal			too late	

Which of the following conclusions can be derived from the information in the diagram?

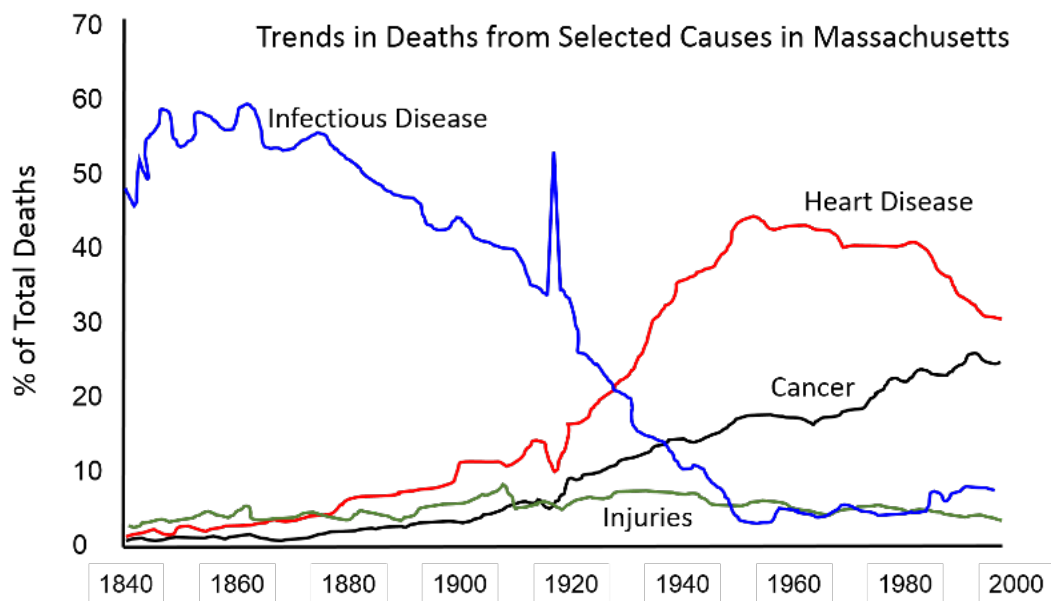
- A. Artificial insemination produces better quality calves than natural mating.
 - B. Artificial insemination has a shorter time frame for conception to occur than natural mating.
 - C. There is a higher chance of conception with natural mating than with artificial insemination.
 - D. Breeders need to know when their cows come into heat to maximise the chance of conception.
19. The diagram shows a simplified version of a human ovum.



Which of the following values is closest to the diameter of this cell?

- A. $7.2 \times 10^{-6}\text{m}$
- B. $1.4 \times 10^{-6}\text{m}$
- C. $7.2 \times 10^{-4}\text{m}$
- D. $1.4 \times 10^{-4}\text{m}$

20. The graph below displays the trends in deaths from selected causes in Massachusetts, USA.



Which of the following statements is the most likely from the information in the graph?

- A. The rise in percentage of total deaths due to cancer around the years 1940 – 1960 is due to an increase in the resistance of bacteria to antibiotics.
- B. The decrease in percentage of total deaths due to infectious disease is due to a decrease in vaccination rates.
- C. The increase in percentage of total deaths due to infectious disease from 1980 to 2000 is due to an increase in poor quality diet and lack of exercise.
- D. The percentage of total deaths due to infectious disease was at its lowest in the 1950s because of the invention of antibiotics and improvements to sanitation.

End of Section I

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Student Number



Barker
College

Biology

Section II Answer Booklet

80 marks.

Attempt Questions 21 - 34

Allow about 2 hours and 25 minutes for this part

Instructions:

- Write your Student Number on *all of your lined pages*.
 - Answer the questions *on your own lined paper*. Clearly indicating which question you are answering.

The spaces provide guidance for the expected length of response.
 - Show all relevant working in questions involving calculations.
 - Extra writing space is provided at the back of this booklet. You will not need to use this space for this Trial examination.
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Student Number

Question 21 (4 marks)

Like mammals, the fruit fly *Drosophila* has a system of sex determination with X and Y chromosomes

The dominant allele of the Bar gene on the X chromosome causes small, narrow eyes, called 'Bar eyes'. The normal eye type is the recessive allele and it causes large, round eyes.

If a male with Bar eyes is crossed with a female with normal eyes, predict the possible offspring.

Show all of your working.

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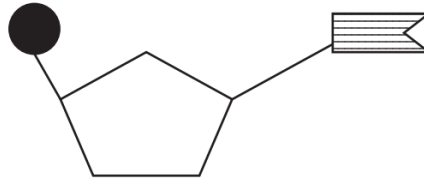
Student Number

Question 22 (3 marks)

RNA molecules consist of long strands of joined nucleotides. Each nucleotide consists of three sub-units.

- (a) Label the THREE sub-units in the diagram of the RNA nucleotide below.

1



- (b) Copy the table below onto your lined writing paper.

Complete the table by describing the role of the two types of RNA in a cell.

2

<i>Type of RNA</i>	<i>Role in a cell</i>
tRNA	
mRNA	

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Student Number

Question 23 (7 marks)

- (a) The human immune system uses several different types of cells to eliminate cells infected by viruses. Name ONE of these cells and outline how it eliminates virally infected cells. **2**

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- (b) Take a new sheet of paper for this question

Use a flowchart to describe the sequence of events that results in the production of antibodies after exposure to a pathogen.

Your flow chart needs to be at least $\frac{3}{4}$ of the page in size.

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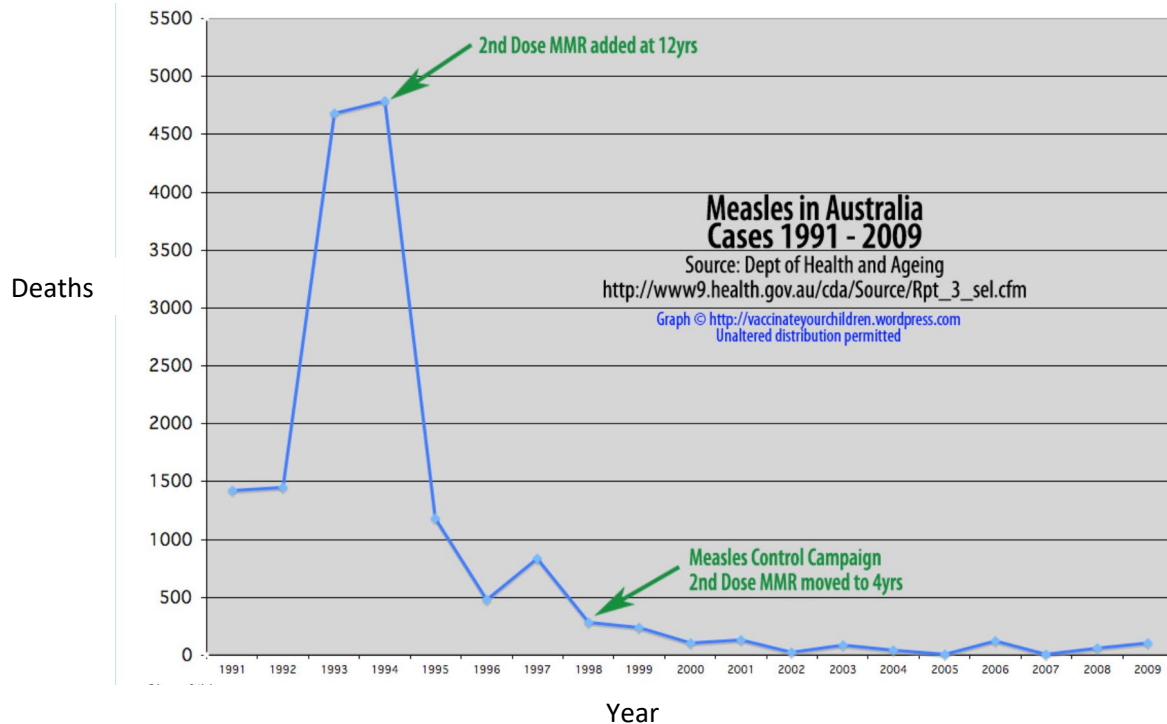
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Student Number

Question 24 (8 marks)

The graph below shows the Deaths from Measles in Australia, 1991 – 2009.
Until 1994, only one dose of the MMR (measles/ mumps/ rubella) vaccine was given between the ages of 12 – 18 months.



File:Graphs measles Australia 1991-2009.jpg 2011, Wikipedia, viewed 30 May 2021,
<https://en.wikipedia.org/wiki/File:Graphs_measles_Australia_1991-2009.jpg>.

- (a) From the early 1990s the *incidence* of measles took a sharp increase. Medical experts made a clear link with the large number of people not bothering to vaccinate their children when it was no longer linked to financial tax incentives. Herd immunity was compromised as a result.

Explain how high vaccination rates provide protection to people who are not vaccinated, such as young children, those unable due to illness or chemotherapy.

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Question 24 continues on page 18

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Student Number

Question 24 (continued)

- (b) With reference to the graph, along with your understanding of adaptive immunity, explain why the recommended vaccination schedule for MMR (Measles /Mumps /Rubella), consists of two doses.

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End of Question 24

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Student Number

Question 25 (3 marks)

Describe the response of a named Australian plant to a specific pathogen.

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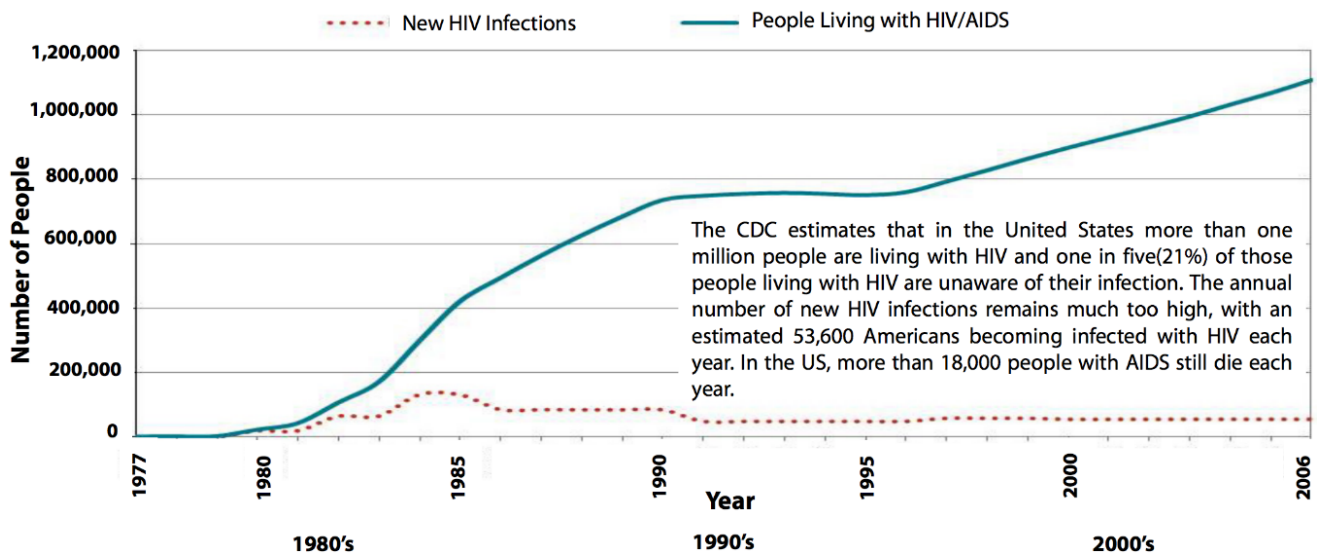
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Question 26 (15 marks)

The two graphs below relate to the numbers of HIV infections in the USA and the number of deaths caused by HIV/AIDS worldwide.

Figure 1: Numbers of new HIV infections and total number of people living with HIV/AIDS in the USA 1977 – 2006.



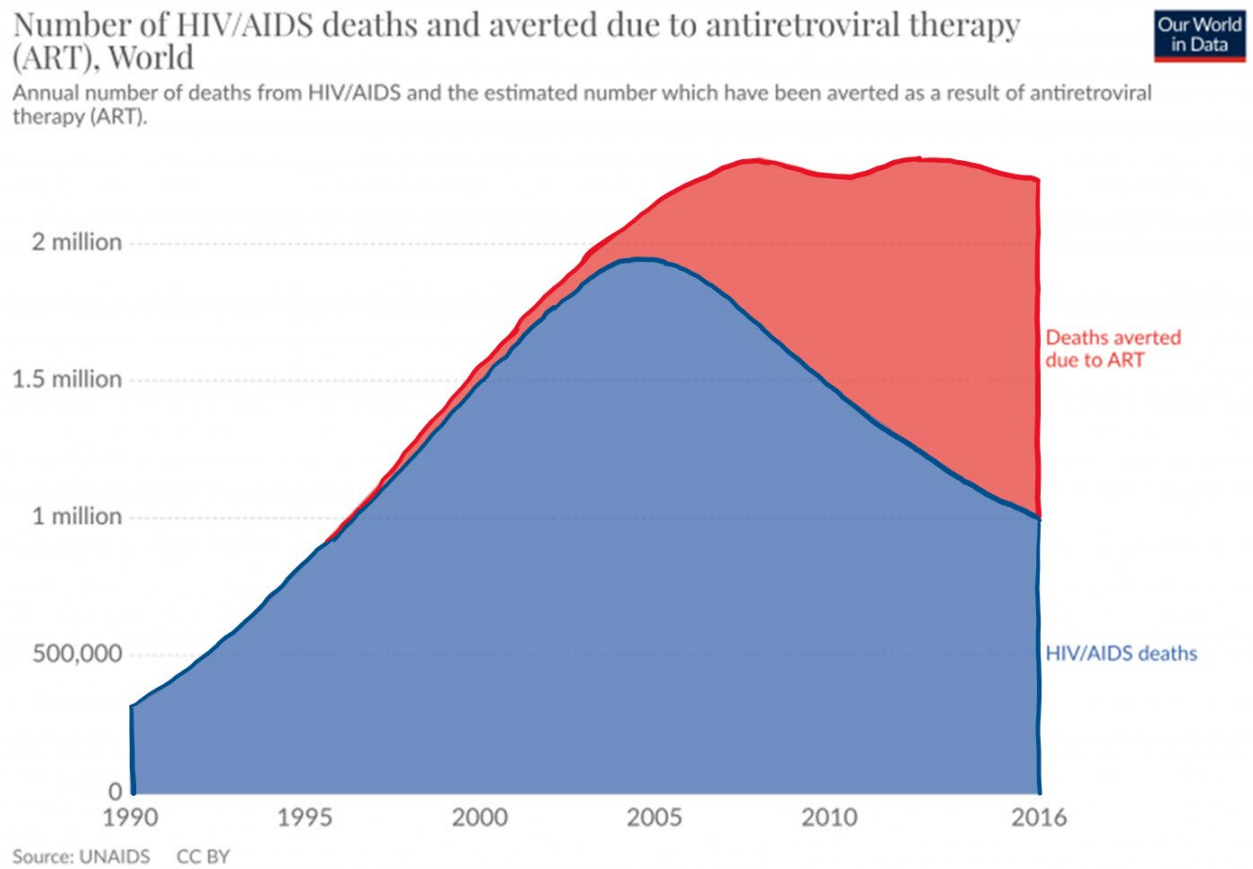
Question 26 continues on page 20

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Student Number

Question 26 (continued)

Figure 2:



(a) Describe the trends shown in Figures 1 and 2.

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Question 26 continues on page 21

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Student Number

Question 26 (continued)

- (b) Antiretroviral medication is a type of anti-viral medicine that impacts the HIV virus. With reference to the stimulus provided, evaluate the effectiveness of antiretroviral medication used to treat people infected with HIV/AIDS.

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Question 26 continues on page 22

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Student Number

Question 26 (continued)

- (c) In addition to the use of pharmaceuticals, there are many other strategies used to prevent the spread of infectious disease. Assess TWO strategies other than the use of pharmaceuticals that can be employed to prevent the spread of a named disease.

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End of Question 26

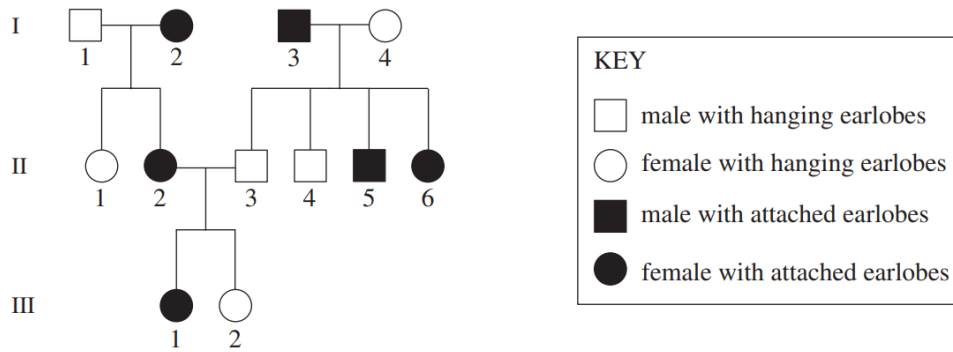
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Student Number

Question 27 (4 marks)

The possession of one of two types of earlobes (hanging or attached) is sometimes used to study the inheritance of characteristics in humans. The pedigree below shows the different types of earlobes present in three generations.

In the first generation, individuals 1 and 4 are heterozygous for the alleles responsible for earlobe type and individuals 2 and 3 are homozygous for these alleles.



Discuss the type of inheritance shown in the pedigree. In your answer, include reference to the alleles using appropriate notation/symbols.

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Student Number

Question 28 (4 marks)

The human menstrual cycle is under the control of four hormones.

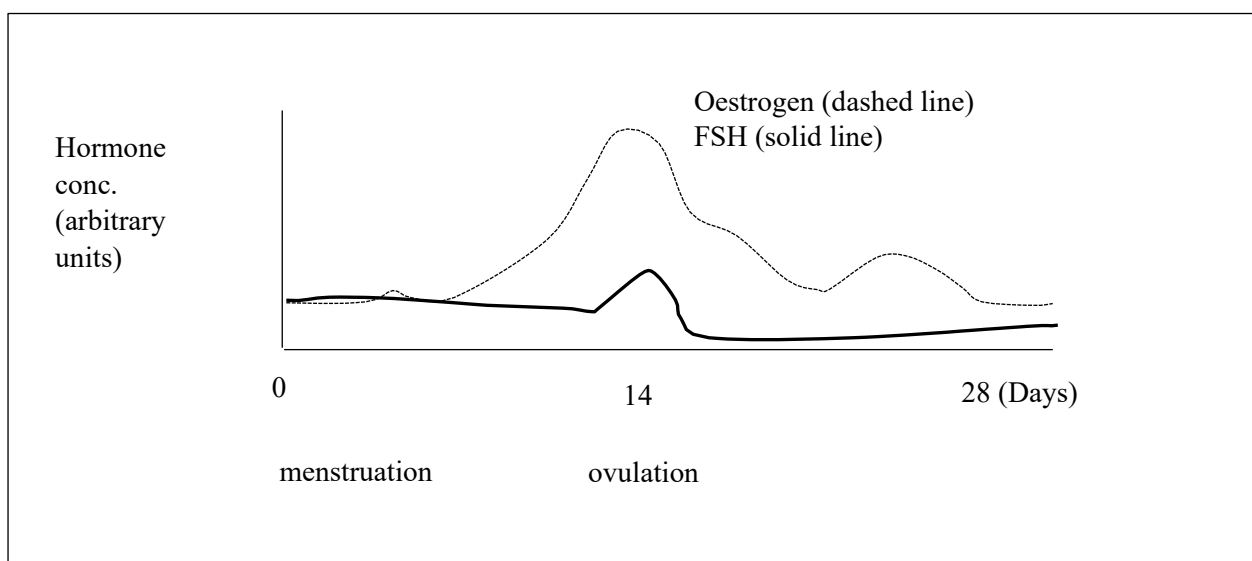
Two, FSH and LH are made by the pituitary gland, and the other two, oestrogen and progesterone, are made within the ovaries.

The information below describes how they act.

The Hormonal Control of the Menstrual Cycle

1. FSH stimulates the 'ripening' of an egg. As it develops it produces oestrogen which, amongst other things 'switches off' FSH production to make sure that only one egg 'ripens' at once.
2. LH levels remain low throughout the cycle, except for a sudden surge which triggers ovulation -the release of the egg, on about the 14th day.
3. The follicle which released the egg then produces progesterone, which targets the lining of the uterus, making it receptive to a fertilised egg should fertilisation occur.
4. After about 22 days of the cycle, if fertilisation hasn't occurred, progesterone and oestrogen levels drop. Once they get below a certain level menstruation is triggered on around the 28th day of the cycle.

The graph below shows how oestrogen and FSH levels change during the cycle.



Question 28 continues on page 25

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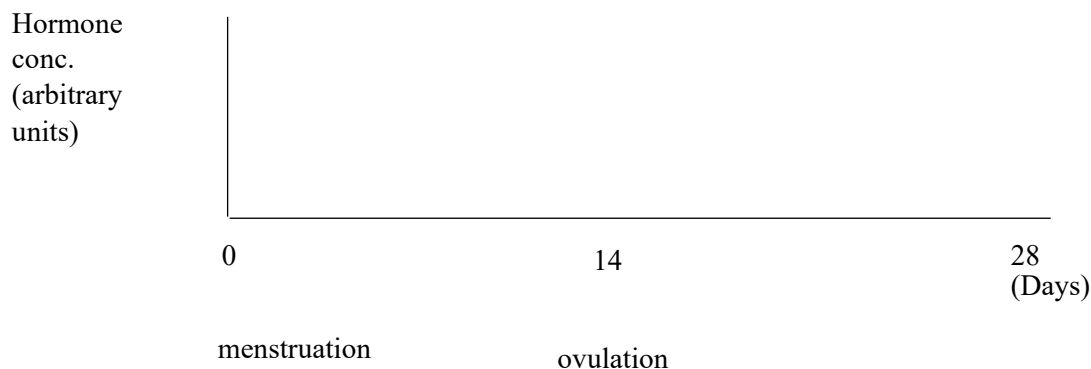
Student Number

Question 28 (continued)

- (a) Copy the axes below onto a new sheet of your lined writing paper.
Make the axes large enough to cover about $\frac{1}{2}$ page.

Draw lines on the graph to show how levels of LH and progesterone also change over the same time.

2



- (b) Outline the role of one named hormone in pregnancy.

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End of Question 28

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Student Number

Question 29 (8 marks) Use a new sheet of lined paper for this question.

Describe, using named examples, two techniques used to manipulate genomes and explain the benefits of these to society and their effect on the Earth's biodiversity.

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Student Number

Question 30 (11 marks) Use a new sheet of lined paper for this question.

Influenza is a disease caused by a group of RNA viruses. There are three types: A, B and C.

Types B and C mutate slowly and do not spread easily, and so are not of serious concern to medical authorities.

Type A viruses have caused several major pandemics.

- (a) Distinguish between a pandemic and an epidemic.

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- (b) Influenza viruses are named according to the variants of two proteins that they possess, known as HG and ND

- HG is used by the virus to attach to a cell so that it can infect it.
- ND is an enzyme that the virus uses to escape from the cell.

Relenza[®] is an anti-viral medication designed to inhibit ND.

Clarify how Relenza[®] reduces the duration of influenza.

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- (c) ND is made from 466 amino acids.

Calculate the minimum number of bases that could make up the gene that codes for it.
Show your working.

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Question 30 continues on page 28

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Student Number

Question 30 (continued)

- (d) The first nine amino acids of the enzyme ND are:

Met Leu Pro Ser Thr Val Gln Thr Leu.

The table below shows the triplets that code for amino acids in the template strand of the DNA extracted from a cell infected with the influenza virus.

<i>Amino acid</i>	<i>Leu</i>	<i>Gln</i>	<i>Met</i>	<i>Pro</i>	<i>Ser</i>	<i>Thr</i>	<i>Val</i>
DNA sequences	AAT	GTT	TAC	GGA	AGA	TGA	CAA
	AAC	GTC		GGG	AGG	TGG	CAG
	GAA			GGT	AGT	TGT	CAT
	GAG			GGC	AGC	TGC	CAC
	GAT				TCA		
	GAC						

Using the information above, identify one of the possible nucleic acid sequences for the influenza virus.

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- (e) If the influenza A virus was to mutate at its second amino acid triplet code, discuss whether the anti-viral medication, Relenza[®] will still be effective.

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End of Question 30

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Student Number

Question 31 (13 marks)

A student produced the following summary of internal and external fertilisation in animals and posted it to a student content-sharing site on the internet. Unfortunately, there were a number of errors in the summary, providing misleading information to students who downloaded the summary for their own study purposes.

- (a) The downloaded summary is shown in the following table.

Discuss TWO errors that the student has made.

3

<i>Characteristic</i>	<i>Differences</i>	
	<i>External fertilisation</i>	<i>Internal fertilisation</i>
Union of gametes	Occurs in water	Usually occurs on land
Relative chance of fertilisation	High	Low
Relative number of offspring	High	Low
Relative breeding frequency	High	Low
Level of parental care	High	Low

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Question 31 continues on page 30

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Question 31 (continued)

Student Number

- (b) What advice would you give to students searching for Biology information on the internet to ensure that the information they were accessing is reliable?

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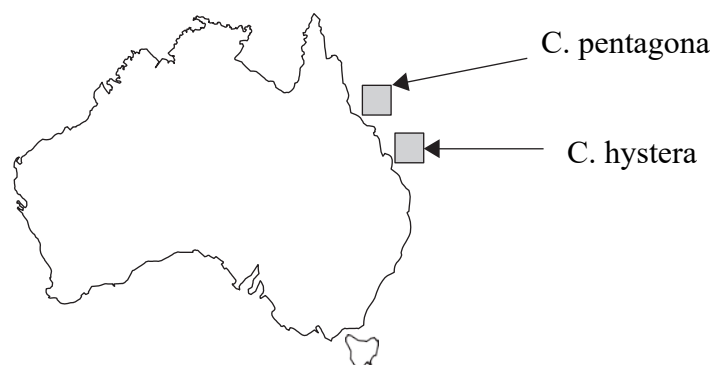
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- (c) Two species of *Cryptasterina* sea stars are found in coastal Queensland. *Cryptasterina pentagona* is found in warmer water further north, while *Cryptasterina hystera* is found further south in cooler water.



One of the phenotypic differences between these two species of sea stars is their method of reproduction. *C. pentagona* reproduces sexually and its sperm and eggs are free-floating in the ocean. *C. hystera* self-fertilises and its fertilised eggs are kept within the sea star until maturity.

The researchers found that one species of *Cryptasterina* has a significantly higher diversity of alleles in its gene pool than the other species.

Take a new sheet of lined paper to answer this question

Explain how each reproduction strategy ensures the continuity of the species. Include in your answer an explanation of which species of *Cryptasterina* would be expected to have the highest allele diversity.

7

Question 31 continues on page 31

